Sree Harisankar S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.95
SKNS PMC VIVEKANANDA VIDYALAYA	Class XII (CBSE)	2023	94.40%
SKNS PMC VIVEKANANDA VIDYALAYA	Class X (CBSE)	2021	91.10%

ACHIEVEMENTS

Kaya AI IIT India Hackathon 2026 (Track 1: Physical AI) -- Team Espada; **shortlisted among top 7 teams (from 1,459 participants)**, advancing to the next stage with Witness, a wearable-assisted construction-site verification system built on Meta smart glasses (ongoing)

Expedia Group Campus Hackathon 2026 -- reached the Finale (Top 20) with the Hotel Review Intelligence Engine; cleared earlier rounds on SQL, algorithmic problem-solving, and competitive programming

PROJECTS

Clinical Risk Dashboard · Python, FastAPI, PostgreSQL, SQLAlchemy/Alembic, React, TypeScript, Docker, scikit-learn, SHAP, MLflow

- Built a **full-stack clinical risk tool**: a **FastAPI + PostgreSQL** service that scores patients, explains each score with **SHAP**, and stores every prediction; a **React/TypeScript** frontend renders the risk, per-feature contributions, and a history trend.
- Showed that headline accuracy (**76%**) hid a **0.59 recall** on positive cases; used **SMOTE and threshold analysis** to raise recall to **0.67**, treating the precision-recall trade-off as a clinical decision.
- Tuned **Random Forest and XGBoost with Optuna** across **80+ MLflow-tracked experiments**; all models sat within cross-validation noise (**ROC-AUC 0.84**).
- Containerized the stack as **three Docker Compose services** (Postgres, API, nginx-served frontend) with healthcheck-gated startup and **Alembic-versioned migrations**; added **Evidently drift monitoring** to flag retraining.

Hotel Review Intelligence Engine | Expedia Group Campus Hackathon 2026, Finale (Top 20)

- Built **aspect-based sentiment analysis** (**16 aspects**, **negation-aware**) on 50K reviews; validated at **F1 0.921** on a **hand-built 45-sentence out-of-corpus holdout** -- chosen after proving the dataset was **template-generated** (recombinations of ~87 sentences), so in-corpus testing would only **measure memorization**
- Applied **Benjamini-Hochberg FDR correction** across **831 statistical tests** on seasonal/segment sentiment gaps; found **0** survived correction despite **94 passing naive p<0.05** -- **reported the null result** instead of the misleading raw one
- Built **intent classification** with **stance resolution** (**prefer/avoid/drop/must-have**) feeding a **multi-signal ranking model** with **hard gates** for dealbreaker constraints
- Designed **evidence disambiguation**: aspects with **<5 mentions score neutral** and get flagged, so "no data" is never mistaken for "bad reviews"

SignSpeak -- Real-Time Sign Language Recognizer · Python, PyTorch, DINOv3, MediaPipe, scikit-learn, FastAPI, OpenCV

- Built an ASL alphabet recognizer using **DINOv3 as a frozen feature extractor** with a lightweight classifier head and **MediaPipe hand-cropping** for consistent input.
- Designed a **leakage-proof evaluation** on a held-out webcam source (different hand, camera, lighting): **~76% honest accuracy** (vs. 99.5% on a naive split), reaching **80-100% on visually-distinct letters**, with precision/recall and a confusion matrix.
- Served the model through a **FastAPI /predict** endpoint with attention-map explainability and a live OpenCV demo.

SKILLS

Languages: Python, C++, SQL

Machine Learning: Scikit-learn, XGBoost, model evaluation (precision/recall, ROC-AUC, confusion matrix), stratified

cross-validation, class imbalance handling (SMOTE), hyperparameter tuning (Optuna)

Deep Learning / Computer Vision: PyTorch, transfer learning (frozen feature extractors), foundation models (DINOv3), Hugging Face Transformers, image classification, OpenCV, attention-map explainability

ML Engineering / MLOps: MLflow (experiment tracking), FastAPI (model serving/REST APIs), SHAP (explainable AI), Evidently (data drift monitoring), Docker

Data & Tools: Pandas, NumPy, Matplotlib, Jupyter, Streamlit, Git/GitHub

RELEVANT COURSES

Introduction to Programming, Artificial Intelligence, Data Structures and Applications, Algorithms and Data Structures Lab, Introduction to embedded systems, Introduction to Probability, Introduction to Statistics, Statistical inference methods in bioengineering.